

EXCEL

VLOOKUP

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INTRODUCTION

- VLOOKUP stands for Vertical Lookup.
- It is a function in Excel and Google Sheets that allows you to search for a value in a vertical column and return a corresponding value from another column in the same row.
- It's used to look up and retrieve data from a specific column in a table based on a matching value in the first column.
- The VLOOKUP function can be a useful tool in Fraud Detection & Technologies to;

1. Flagging Suspicious Patterns in Account Activity
2. Monitoring Unusual Access to Sensitive Data
3. Verifying User Identifications Against Known Data
4. Analyzing Behavioral Trends

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SYNTAX OF VLOOKUP

- VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

1. lookup_value – The value you want to search for in the first column of the table (can be a number, text, or reference).
2. table_array – The range of cells that contains the data you want to search through. It includes both the column with the lookup value and the column(s) from which you want to retrieve data.
3. col_index_num – The column number in the table from which to return the value. The first column in the range is 1, the second is 2, and so on.
4. [range_lookup] (optional) – This is a TRUE or FALSE value that defines whether you want an approximate match (TRUE) or an exact match (FALSE).

i. TRUE – Approximate match (default).

ii. FALSE – Exact match.

COMMON USES OF VLOOKUP

EXACT MATCH (FALSE)

- Looks for an exact match, and if it doesn't find one, it returns an error (typically #N/A).
- =VLOOKUP("malicious", A2:B10, 2, FALSE)
- This formula looks for the exact match of "malicious" in the first column of the range A2:A10 and returns the value from the second column in the same row (B2:B10). If "malicious" isn't found exactly, it returns an error.

APPROXIMATE MATCH (TRUE)

- Looks for the closest match (less than or equal to the lookup value), which is useful for finding ranges or thresholds.
- =VLOOKUP(75, A2:B10, 2, TRUE)
- In this case, VLOOKUP looks for the closest value less than or equal to 75 in the first column of A2:A10

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LIMITATIONS OF VLOOKUP

1. Left-to-Right Lookup – VLOOKUP can only look up values in columns to the right of the lookup column. It cannot search for a value to the left of the lookup column.
2. Data Range Limitation – VLOOKUP requires the lookup value to be in the first column of the range. You cannot look for a value in a column that is not the first column of the range.
3. No Exact Match for Sorted Data – When performing an approximate match (using TRUE for the range_lookup), the data must be sorted in ascending order. If it's not sorted, VLOOKUP may give incorrect results.
4. Performance Issues with Large Data
5. Returns Only One Value

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HOW TO OVER COME THE LIMITATIONS OF VLOOKUP

- ❑ =INDEX(C1:C10, MATCH(A1, B1:B10, 0))
- This method allows you to lookup values in any column and return values from any other column, even if the lookup column is not the first column.

- ❑ =FILTER(C1:C10, B1:B10 = A1)
- Returns all matching values.
- ❑ =IFERROR(VLOOKUP(A1, B1:C10, 2, FALSE), "Not Found")
- Handle errors gracefully
- ❑ =TEXTJOIN(", ", TRUE, IF(B1:B10 = A1, C1:C10, ""))
- This formula will return all matching values from column C where column B matches A1, separated by commas.

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